INFINITESIMAL AUTOMORPHISMS AND TANGENT FUNCTORS UNDER (RS*)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3526-3552 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 3554-3628, the complete author proof (73 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_artin.json (pb=2412, stmt 3526-3552, proof 3554-3628).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 06I6: sources/stacks-artin/pinned-original-source.tex, source lines 3628–3628. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$. Assume $\mathcal{X}$ satisfies condition (RS*). Let A be an S-algebra and let x be an object of $\mathcal{X}$ over* $\mathrm{Spec}(A)$.

(1) *There exists an A-linear functor $Inf_x : Mod_A \to Mod_A$ such that given a deformation situation $(x, A' \to A)$ and a lift x' there is an isomorphism $Inf_x(I) \to Inf(x'/x)$ where $I = \mathrm{Ker}(A' \to A)$.*

(2) *There exists an A-linear functor $T_x : Mod_A \to Mod_A$ such that*
   (a) *given M in Mod_A there is a bijection $T_x(M) \to Lift(x, A[M])$,*
   (b) *given a deformation situation $(x, A' \to A)$ there is an action*

$$T_x(I) \times Lift(x, A') \to Lift(x, A')$$

where $I = \mathrm{Ker}(A' \to A)$. It is simply transitive if $Lift(x, A') \neq \emptyset$.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We define Inf_x as the functor

$$\mathrm{Mod}_A \longrightarrow \mathit{Sets}, \quad M \longrightarrow \mathrm{Inf}(x'_M/x) = \mathrm{Lift}(\mathrm{id}_x, A[M])$$

mapping M to the group of infinitesimal automorphisms of the trivial deformation x'_M of x to $\mathrm{Spec}(A[M])$ or equivalently the group of lifts of id_x in $Aut_{\mathcal{X}}(x'_M)$. We define T_x as the functor

$$\mathrm{Mod}_A \longrightarrow \mathit{Sets}, \quad M \longrightarrow \mathrm{Lift}(x, A[M])$$

of isomorphism classes of infinitesimal deformations of x to $\mathrm{Spec}(A[M])$. We apply Formal Deformation Theory, Lemma 06I6 to Inf_x and T_x. This lemma is applicable, since (RS*) tells us that

$$\mathit{Lift}(x, A[M \times N]) = \mathit{Lift}(x, A[M]) \times \mathit{Lift}(x, A[N])$$

as categories (and trivial deformations match up too).

Let $(x, A' \to A)$ be a deformation situation. Consider the ring map $g : A' \times_A A' \to A[I]$ defined by the rule $g(a_1, a_2) = \overline{a_1} \oplus a_2 - a_1$. There is an isomorphism

$$A' \times_A A' \longrightarrow A' \times_A A[I]$$

given by $(a_1, a_2) \mapsto (a_1, g(a_1, a_2))$. This isomorphism commutes with the projections to A' on the first factor, and hence with the projections to A. Thus applying (RS*) twice we find equivalences of categories

$$\begin{aligned} \mathit{Lift}(x, A') \times \mathit{Lift}(x, A') &= \mathit{Lift}(x, A' \times_A A') \\ &= \mathit{Lift}(x, A' \times_A A[I]) \\ &= \mathit{Lift}(x, A') \times \mathit{Lift}(x, A[I]) \end{aligned}$$

Using these maps and projection onto the last factor of the last product we see that we obtain "difference maps"

$$\mathrm{Inf}(x'/x) \times \mathrm{Inf}(x'/x) \longrightarrow \mathrm{Inf}_x(I) \quad \text{and} \quad \mathrm{Lift}(x, A') \times \mathrm{Lift}(x, A') \longrightarrow T_x(I)$$

These difference maps satisfy the transitivity rule "$(x'_1 - x'_2) + (x'_2 - x'_3) = x'_1 - x'_3$" because

$$\begin{array}{ccc} A' \times_A A' \times_A A' & \xrightarrow{(a_1,a_2,a_3) \mapsto (g(a_1,a_2), g(a_2,a_3))} & A[I] \times_A A[I] = A[I \times I] \\ & \searrow^{(a_1,a_2,a_3) \mapsto g(a_1,a_3)} & \downarrow + \\ & & A[I] \end{array}$$

is commutative. Inverting the string of equivalences above we obtain an action which is free and transitive provided $\mathrm{Inf}(x'/x)$, resp. $\mathrm{Lift}(x, A')$ is nonempty. Note that $\mathrm{Inf}(x'/x)$ is always nonempty as it is a group. □